

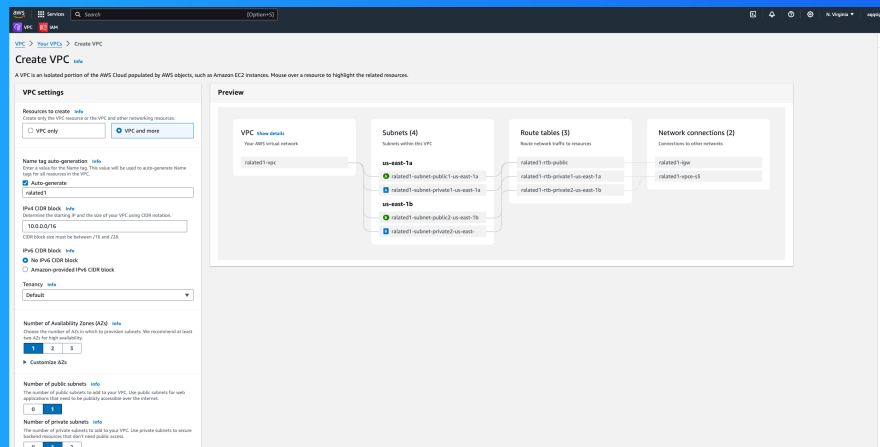


NextWork.org

Launching VPC Resources



alexten3517@gmail.com





Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a AWS foundation networking service that helps us create our own networks and configure the security, traffic rules and connectivity with the internet.

How I used Amazon VPC in this project

I used Amazon VPC to create my own VPC with different resources/components(e.g security groups,network ACL's).

One thing I didn't expect in this project was...

I did not expect the project to really teach me the importance of a private VPC and how to launch a EC2 resource while explaining the security measure one must take to keep privacy as a major objective.

This project took me...

This project took me 2 hours with research and live demo time.



Setting Up Direct VM Access

Directly accessing an EC2 virtual machine means "logging into the EC2 instance", instead of just managing it at a higher level with the AWS Management Console. This includes installing software and changing my EC2 instance's configurations.

SSH is a key method for directly accessing a VM

SSH means Secure Shell and its both a protocol and a traffic type. It is a protocol that matches keypair and enables direct VM access and once a connection is set up, its is a traffic type that encrypts comm.data being transferred.

To enable direct access, I set up key pairs

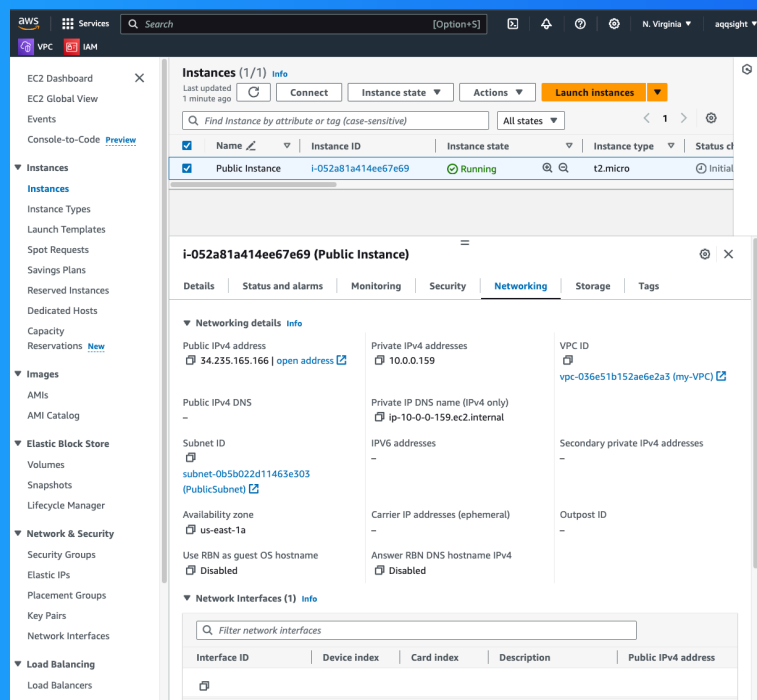
An AWS key pair is a pair of cryptographic keys used to authenticate and access your (EC2) instances. While key pairs are primarily used for secure SSH access, they can also be used for other authentication such as API calls.

My private key's file format was set up as .pem, which is file format widely accepted many servers. A format most servers are able to read/use.



Launching a public server

I had to change my EC2 instance's networking settings by changing the VPC and subnet my EC2 was going to be launched in. I updated both my public VPC and my public subnet while also utilizing my public security group.

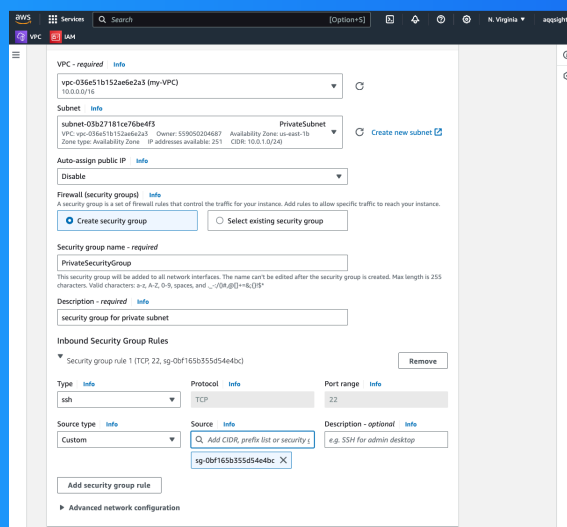




Launching a private server

My private server has its own dedicated security group because the public security group allows in all HTTP traffic which would leave our private server more vulnerable to attacks/risks.

My private server's security group, which means only SSH traffic coming from resources associated with that security group would be allowed.



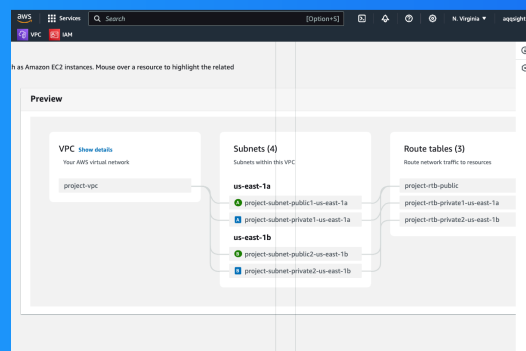


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time I used the VPC and more option which gives me a resource map to use when creating the VPC and all its components.

A VPC resource map is a visual representation that maps out my vpc and all the components that are and the connections between them. It shows me the highlighted connections.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because VPC are already isolated from each other.



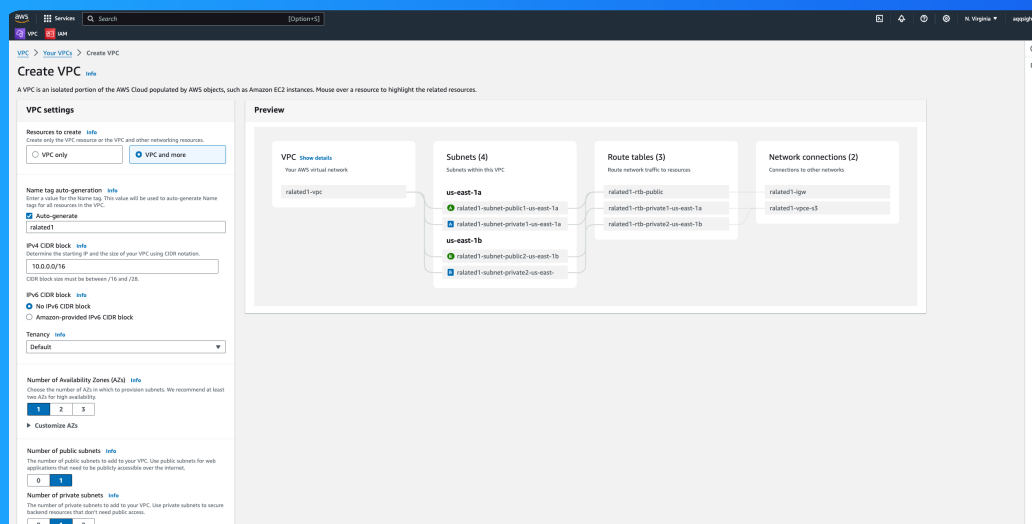


Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: either none or 1 per each availability zone for my VPC. This is because it is best practice (improves redundancy and high availability) to have at least subnet/AZ.

The set up page also offered to create NAT gateways, typically a NAT Gateway or VPN gateway, allows you to access resources within a private VPC from the public internet. This means you can communicate with EC2 instances.





NextWork.org

Everyone should be in a job they love.

Check out nextwork.org for
more projects

